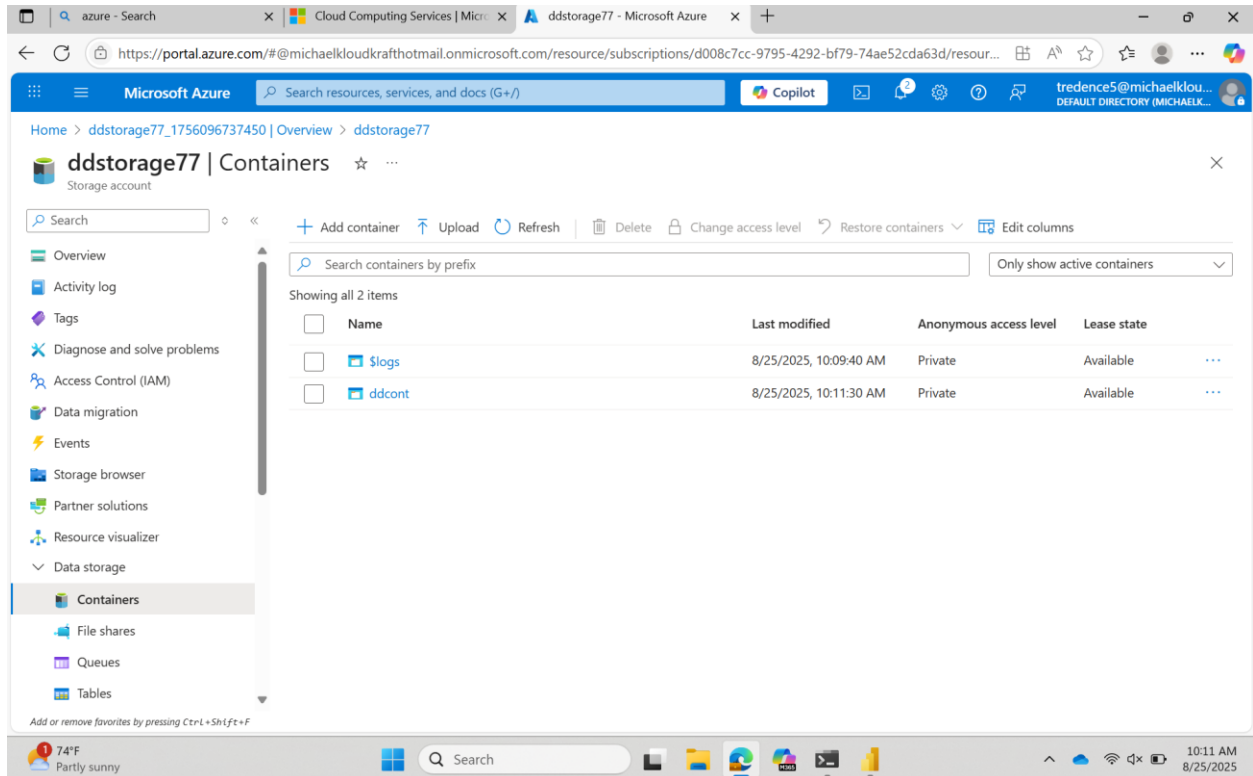


Devanshu deo
Capstone Project – 1
IOT-Enabled Retail Analytics on Azure Databricks
1) Creating a storage account



Creating an access connector for databricks

Microsoft Azure Search resources, services, and docs (G+/J) Copilot tredence5@michaelklo... DEFAULT DIRECTORY (MICHAELK...

Home > AccessConnectorCreate-20250825103312 | Overview Deployment

Search x << Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : AccessConnectorCreate-20250... Start time : 8/25/2025, 10:34:33 AM
Subscription : LabsKraft Correlation ID : 933346c5-0629-4cd1-ac48-27...
Resource group : LabsKraft2027

> Deployment details

Next steps

Go to resource

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

Microsoft Defender for Cloud
Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials
[Start learning today >](#)

Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
[Find an Azure expert >](#)

Add or remove favorites by pressing Ctrl+Shift+F

Creating the external location for unity catalog

Microsoft Azure databricks Search data, notebooks, recents, and more... CTRL + P tred_d2 v 1

+ New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Catalog Explorer > External Locations >

ddext

Test connection

Overview Permissions Browse Workspaces

Description

Add description

About this external location

Owner Tredence LabsKraft5

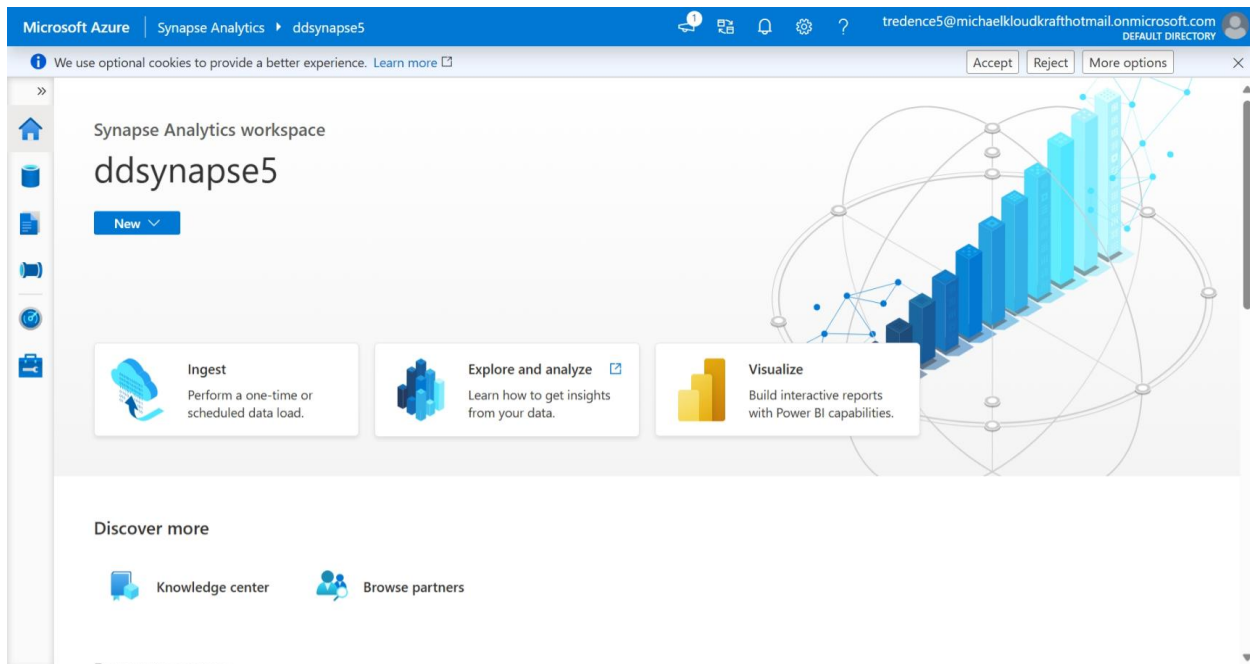
Fallback mode: ☐

Credential	ddext_azuremanagedidentity_1756099151051
URL	abfss://ddcont5@ddstorage77.dfs.core.windows.net/
Limit to read-only use	Disabled

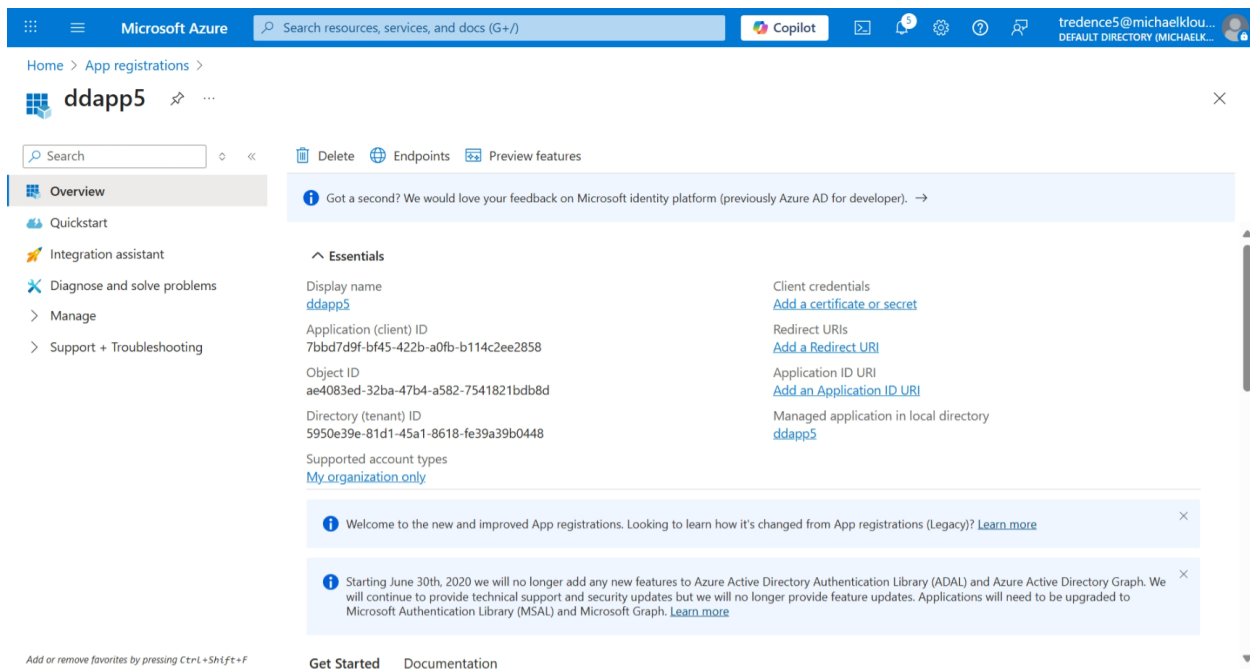
The screenshot shows the Databricks Catalog Explorer interface. On the left, a sidebar lists various components like Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Job Runs, Data Ingestion, and AI/ML. The main area displays the 'ddcatalog5' catalog under 'My organization'. It lists several schemas: default, information_schema, tredence_02, aman_cata, capstone03, diksha-first-catalog, diksha-second-catalog, haseebscatalog, hetul_catalog, hetul_new_catalog, hetul_project01_catalog, retail_analytics_catalog, and Delta Shares Received. The 'ddcatalog5' catalog is selected, and its details are shown on the right. The 'Overview' tab is active, showing a description, a filter for schemas (2 schemas), and a table of schemas. The table has columns for Name, Owner, and Created at. The schemas listed are 'default' (owned by tredence5@michaelkloudkrafth...) and 'information_schema' (owned by System user). Below the table, there is an 'About this catalog' section showing the owner as Tredence LabsKraft5 and a 'Tags' section with an 'Add tags' button.

Deploying a new azure synapse analytics

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the Copilot button. The main content area displays the 'Overview' page for the deployment 'Microsoft.Azure.SynapseAnalytics-20250825105440'. The deployment is marked as 'Your deployment is complete' with a green checkmark. The deployment details show the name, subscription (LabsKraft), resource group (LabsKraft2027), start time (8/25/2025, 11:05:08 AM), and correlation ID (fd691e61-966a-45ca-a220-e83...). The 'Next steps' section includes a 'Go to resource group' button. The 'Give feedback' section has a link to 'Tell us about your experience with deployment'. On the right, there are three sections: 'Cost management' (Get notified to stay within your budget and prevent unexpected charges on your bill. Set up cost alerts >), 'Microsoft Defender for Cloud' (Secure your apps and infrastructure. Go to Microsoft Defender for Cloud >), and 'Free Microsoft tutorials' (Start learning today >). The 'Work with an expert' section mentions Azure experts as service provider partners who can help manage your assets on Azure and be your first line of support. Find an Azure expert >.



Creating a new app registration



Creating three schemas in the medallion architecture

Microsoft Azure databricks

Search data, notebooks, recent, and more... CTRL + P

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Untitled Notebook 2025-08-25 11:27:13

File Edit View Run Help Python Tabs: ON Last edit was 1 minute ago Run all kraft100 Schedule Share

```
%sql
USE CATALOG ddcatalog5;

CREATE SCHEMA IF NOT EXISTS bronze;
CREATE SCHEMA IF NOT EXISTS silver;
CREATE SCHEMA IF NOT EXISTS gold;
```

11:29 AM (6s) 1

```
# Parameters
storage_account = "ddstorage77"
container = "ddcont5"
file_path = "retail_iot_data.csv"

# Credentials - Replace with actual values (or keep them in a secret scope)
client_id = "7bbd7d9f-bf45-422b-a0fb-b114c2ee2858"
tenant_id = "5950e39e-81d1-45a1-8618-fe39a39b0448"
client_secret = "Bjm8Q~E6wahK1OPQhzRTzQzYyMr_WqSfxCFmccC8"

# -----
# Configurations for ABFS access (no mount needed in UC)
# -----
```

1 minute ago (10s) 2 Python

Code for bronze layer:

Microsoft Azure databricks

Search data, notebooks, recent, and more... CTRL + P

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Untitled Notebook 2025-08-25 11:27:13

File Edit View Run Help Python Tabs: ON Last edit was 1 minute ago Run all kraft100 Schedule Share

```
.csv(abfs_path)

)

print(f"Loaded {df_bronze.count()} rows from {abfs_path}")
display(df_bronze.limit(5))
```

Generate (Ctrl + I)

(9) Spark Jobs

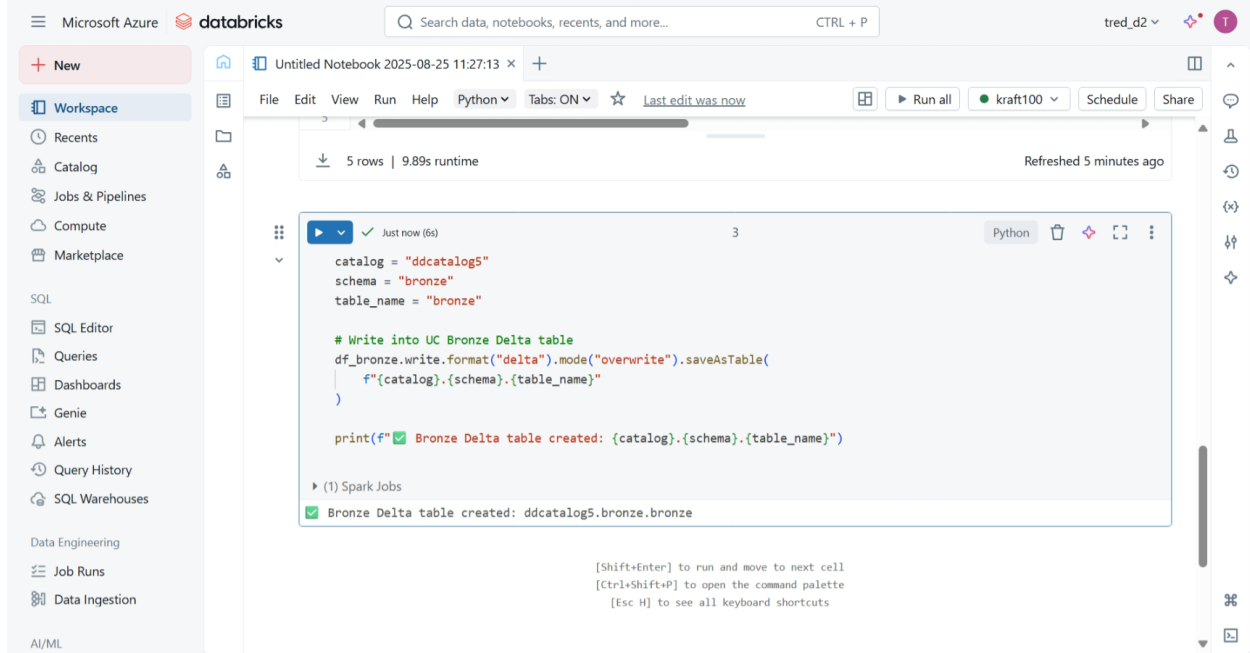
df_bronze: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ..., 12 more fields]

Loaded 1000 rows from abfss://ddcont5@ddstorage77.dfs.core.windows.net/retail_iot_data.csv

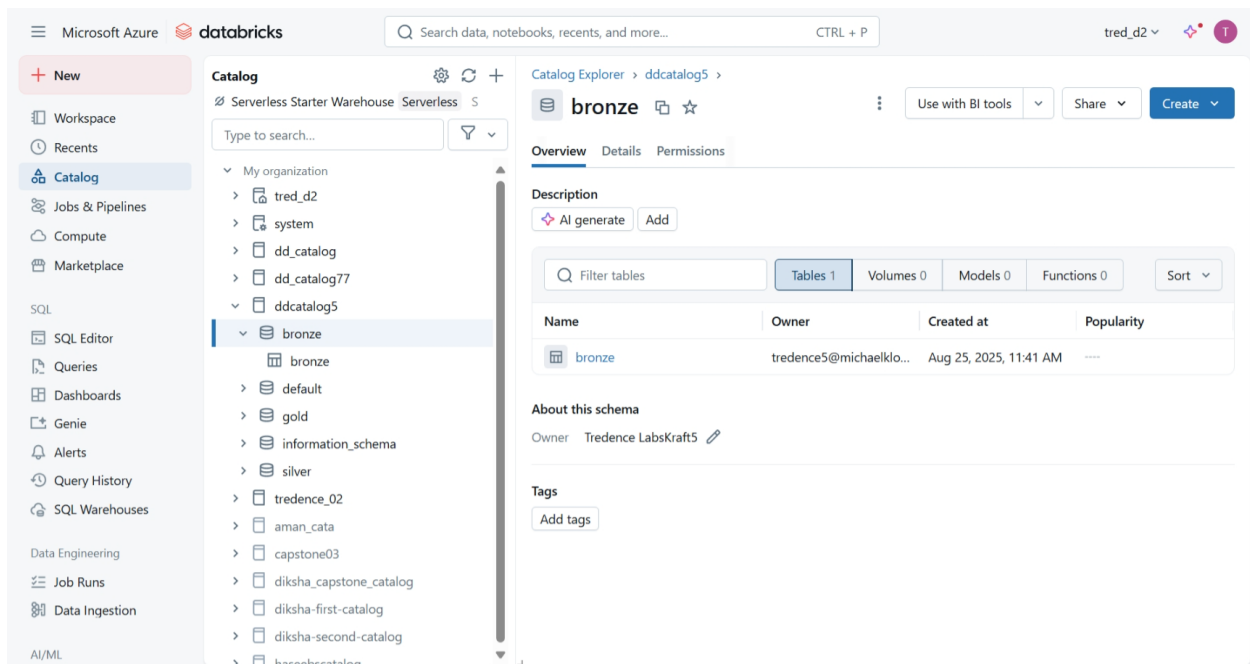
	OrderID	CustomerID	Region	StoreID	Product	Quantity
1	db07ec0d-e8e7-408a-846e-c2d330484d...	CUST-2824	East	Store-009	Mobile	2
2	93d89258-6ef2-4bb8-9a71-cb590cb55f85	CUST-7912	East	Store-001	Laptop	2
3	f97636e9-3dab-48ad-aad4-0eca034d3111	CUST-9928	South	Store-008	Tablet	5
4	f20eae76-3e09-4580-8429-af1d6b76209a	CUST-3547	West	Store-011	Laptop	1
5	...	...	...	...	...	...

5 rows | 9.89s runtime

Refreshed 1 minute ago



Bronze table that we just created in the schema with the appropriate columns



Creating the silver layer

Microsoft Azure databricks

Search data, notebooks, recent, and more... CTRL + P

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Untitled Notebook 2025-08-25 11:27:13

File Edit View Run Help Python Tabs: ON Last edit was 1 minute ago Run all kraft100 Schedule Share

```
import pyspark.sql.functions as F
bronze_df = spark.table("ddcatalog5.bronze.bronze")
silver_df = (bronze_df
    .dropDuplicates(["OrderID"])
    .withColumn("OrderTime", F.to_timestamp("OrderTime"))
    .withColumn("Region", F.upper(F.col("Region")))
    .withColumn("AnomalyType", F.coalesce(F.col("AnomalyType"), F.lit("None")))
    .withColumn("IsTempAnomaly",
        F.when((F.col("Temperature") < 0) | (F.col("Temperature") > 35), 1).otherwise(0))
    )
silver_df.write.format("delta").mode("overwrite").saveAsTable(
    "ddcatalog5.silver.retail_cleaned")
print("Silver table created: ddcatalog5.silver.retail_cleaned")
```

(3) Spark Jobs

- bronze_df: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 12 more fields]
- silver_df: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 13 more fields]

Silver table created: ddcatalog5.silver.retail_cleaned

Creating different KPI's for the gold layer

Microsoft Azure databricks

Search data, notebooks, recent, and more... CTRL + P

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Untitled Notebook 2025-08-25 11:27:13

File Edit View Run Help Python Tabs: ON Last edit was 1 minute ago Run all kraft100 Schedule Share

```
#gold layer - daily/monthly sales per region
import pyspark.sql.functions as F

sdf = spark.table("ddcatalog5.silver.retail_cleaned")

monthly_sales = (sdf
    .groupBy("Region", F.date_trunc("month", "OrderTime").alias("Month"))
    .agg(F.sum("SalesAmount").alias("TotalSales"),
        F.sum("Quantity").alias("UnitsSold"))
    )

monthly_sales.write.format("delta").mode("overwrite").saveAsTable(
    "ddcatalog5.gold.monthly_sales")
```

(3) Spark Jobs

- monthly_sales: pyspark.sql.connect.dataframe.DataFrame = [Region: string, Month: timestamp ... 2 more fields]
- sdf: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 13 more fields]

Microsoft Azure databricks

Search data, notebooks, reents, and more... CTRL + P

tred_d2

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Untitled Notebook 2025-08-25 11:27:13

File Edit View Run Help Python Tabs: ON ☆ Last edit was 7 minutes ago

Run all kraft100 Schedule Share

01:42 PM (5s) 10

```
anomaly_trend = (sdf
    .groupBy("StoreID", F.to_date("OrderTime").alias("Day"))
    .agg(F.sum("AnomalyFlag").alias("SystemAnomalies"),
        F.sum("IsTempAnomaly").alias("TempAnomalies"),
        F.sum("SalesAmount").alias("Sales"))
)

anomaly_trend.write.format("delta").mode("overwrite").saveAsTable(
    "ddcatalog5.gold.anomaly_trend"
)
```

(2) Spark Jobs

anomaly_trend: pyspark.sql.connect.dataframe.DataFrame = [StoreID: string, Day: date ... 3 more fields]

01:43 PM (2s) 11

```
display(anomaly_trend)
```

(2) Spark Jobs

Table +

StoreID	Day	SystemAnomalies	TempAnomalies	1.2 Sales
1111	2025-08-25	1	1	1.2

Microsoft Azure databricks

Search data, notebooks, reents, and more... CTRL + P

tred_d2

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Untitled Notebook 2025-08-25 11:27:13

File Edit View Run Help Python Tabs: ON ☆ Last edit was 8 minutes ago

Run all kraft100 Schedule Share

02:13 PM (10s) 12

```
#gold layer kpi3 - impact of device failure

impact = (sdf
    .withColumn("AnomalyDetected", (F.col("AnomalyFlag") + F.col("IsTempAnomaly")) > 0)
    .groupBy("AnomalyDetected")
    .agg(F.sum("SalesAmount").alias("TotalSales"),
        F.countDistinct("OrderID").alias("Orders"))
)

impact.write.format("delta").mode("overwrite").saveAsTable(
    "ddcatalog5.gold.anomaly_impact"
)
```

(3) Spark Jobs

impact: pyspark.sql.connect.dataframe.DataFrame = [AnomalyDetected: boolean, TotalSales: double ... 1 more field]

02:16 PM (9s) 13

```
#gold layer kpi4 - store into tiers

store_perf = (sdf
    .groupBy("StoreID")
    .agg(F.sum("SalesAmount").alias("TotalSales"))
    .withColumn("Tier",
```


Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

tredecence5@michaelklo...
DEFAULT DIRECTORY (MICHAELK...

Home > Storage accounts > ddstorage77 | Access Control (IAM) >

Add role assignment ...

RoleMembersConditionsReview + assign

Showing a filtered list of roles because your permissions include a condition. [Learn more](#)
[View my access](#)

Selected role

Storage Blob Data Reader

Assign access to

User, group, or service principal

Managed identity

Members

+ Select members

Name	Object ID	Type
ddsnapse5	813da8f3-3239-4b35-b313-b57f80cad6...	Synapse workspace ⓘ

Description

Optional

Review + assign

Previous

Next

Feedback

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

tredecence5@michaelklo...
DEFAULT DIRECTORY (MICHAELK...

Home > Storage accounts > ddstorage77

ddstorage77 | Access Control (IAM) ☆ ...

Storage account

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Partner solutions

Resource visualizer

Data storage

Containers

File shares

Queues

Tables

+ Add

Download role assignments

Edit columns

Refresh

Delete

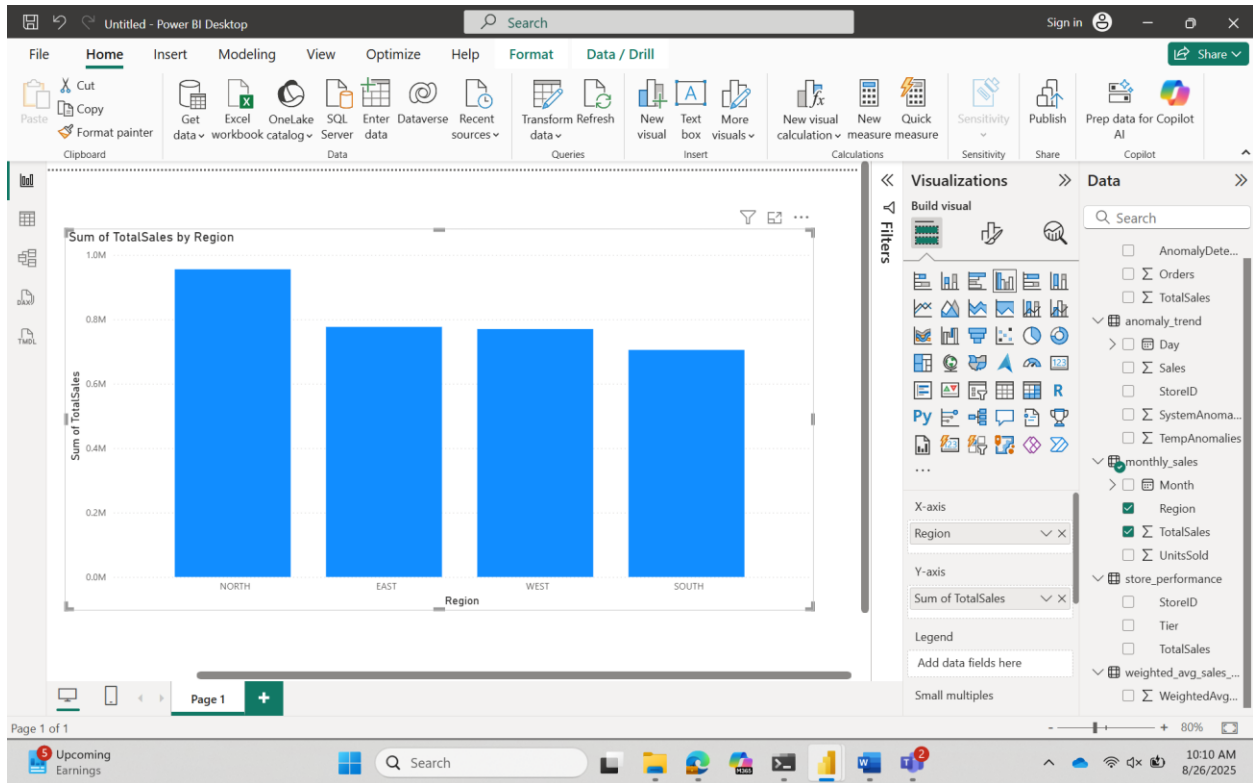
Feedback

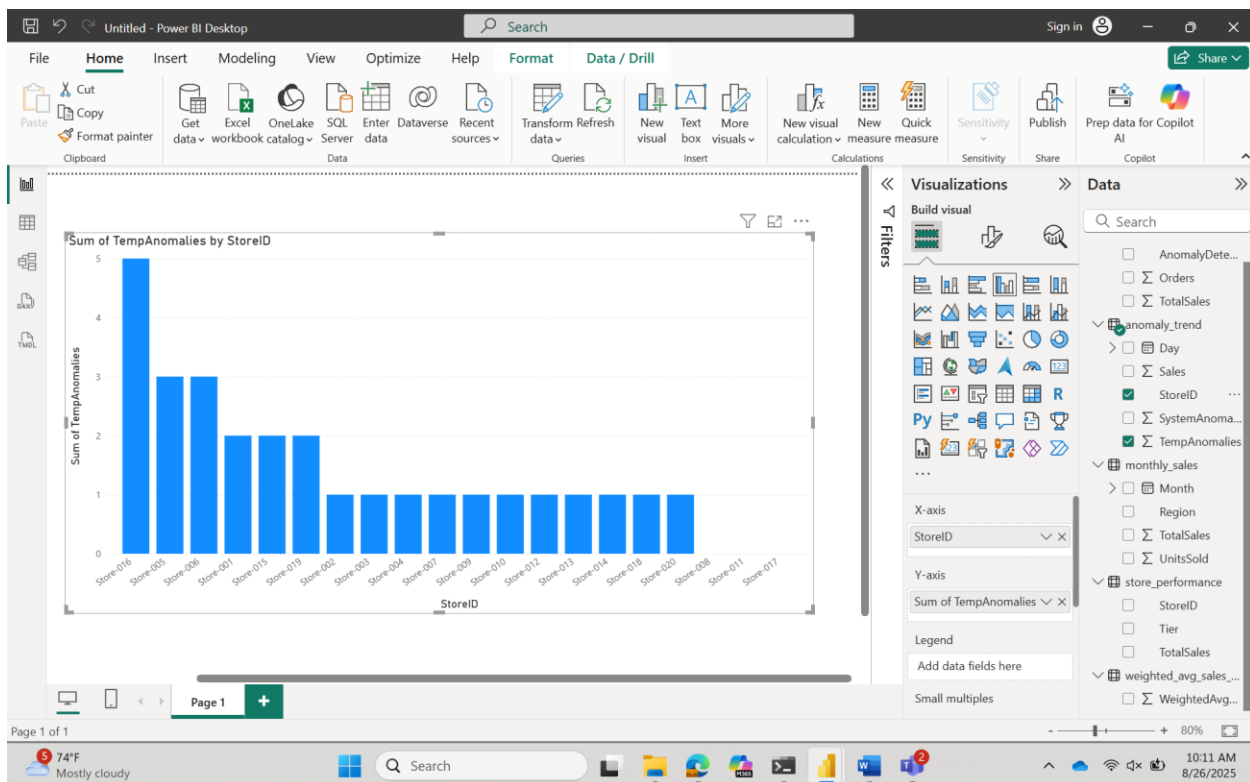
☐	713b74b4-6e77-44...	Managed identity	Contributor	Resource group (Inher...	None
Monitoring Contributor (1)					
☐	SherlockShinde ff837afc-8713-4ff0-...	User	Monitoring Contributor	Subscription (Inherited)	None
Storage Blob Data Contributor (3)					
☐	ddconnect aaf6ca74-bd17-43e...	Managed identity	Storage Blob Data Contrib...	This resource	Add
☐	ddsnapse5 91e42759-1c43-4d...	Managed identity	Storage Blob Data Contrib...	This resource	Add
☐	Tredence LabsKraft5 4a8e9882-f32a-48d...	User	Storage Blob Data Contrib...	This resource	Add
Storage Blob Data Reader (1)					
☐	ddsnapse5 91e42759-1c43-4d...	Managed identity	Storage Blob Data Reader	This resource	Add
User Access Administrator (1)					
☐	Michael A e1fb5008-9e9f-4e28...	User	User Access Administrator	Root (Inherited)	None

Showing 1 - 38 of 38 results.

Add or remove favorites by pressing Ctrl+Shift+F

Power BI dashboards





Optimization of the queries

```
%sql
OPTIMIZE ddcatalog5.gold.monthly_sales ZORDER BY (Month);

OPTIMIZE ddcatalog5.gold.anomaly_trend ZORDER BY (StoreID, Day);
OPTIMIZE ddcatalog5.gold.anomaly_impact ZORDER BY (AnomalyDetected);
OPTIMIZE ddcatalog5.gold.store_performance ZORDER BY (TotalSales);
OPTIMIZE ddcatalog5.gold.weighted_avg_sales_vs_anomalies;

OPTIMIZE ddcatalog5.gold.monthly_sales;
OPTIMIZE ddcatalog5.gold.anomaly_trend;
OPTIMIZE ddcatalog5.gold.anomaly_impact;
OPTIMIZE ddcatalog5.gold.store_performance;
OPTIMIZE ddcatalog5.gold.weighted_avg_sales_vs_anomalies;
```

Using Event Hub as a Data Source

The screenshot shows the 'Create Namespace' blade in the Microsoft Azure portal. The breadcrumb navigation is 'Home > Event Hubs >'. The page title is 'Create Namespace' with a sub-header 'Event Hubs'. There are tabs for 'Basics', 'Advanced', 'Networking', 'Tags', and 'Review + create'. The 'Basics' tab is selected.

Project Details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Instance Details

Enter required settings for this namespace, including a price tier and configuring the number of units (capacity).

Namespace name * ☒ .servicebus.windows.net

Region *
 The region selected supports Availability zones. Your namespace will have Availability Zones enabled. [Learn more](#).

Buttons: [Review + create](#), [< Previous](#), [Next: Advanced >](#)

The screenshot shows the 'ddevent7 | Overview' blade in the Microsoft Azure portal. The breadcrumb navigation is 'Home >'. The page title is 'ddevent7 | Overview' with a sub-header 'Deployment'. There is a search bar and action buttons: 'Delete', 'Cancel', 'Redeploy', 'Download', and 'Refresh'.

Overview

- Inputs
- Outputs
- Template

Your deployment is complete

Deployment name : ddevent7
Subscription : LabsKraft
Resource group : LabsKraft2027

Start time : 8/26/2025, 10:23:10 AM
Correlation ID : 13c65d49-4ec6-46a4-8769-ca...

Deployment details

Next steps

[Go to resource](#)

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill. [Set up cost alerts >](#)

Microsoft Defender for Cloud

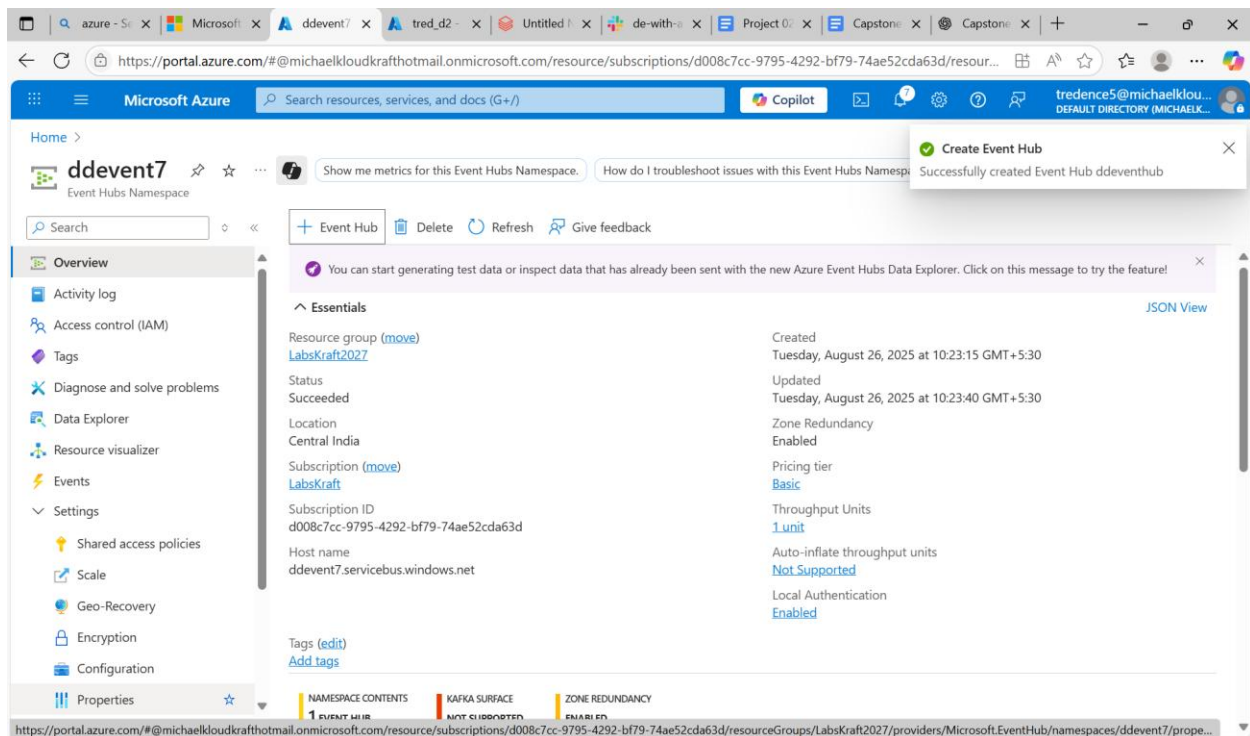
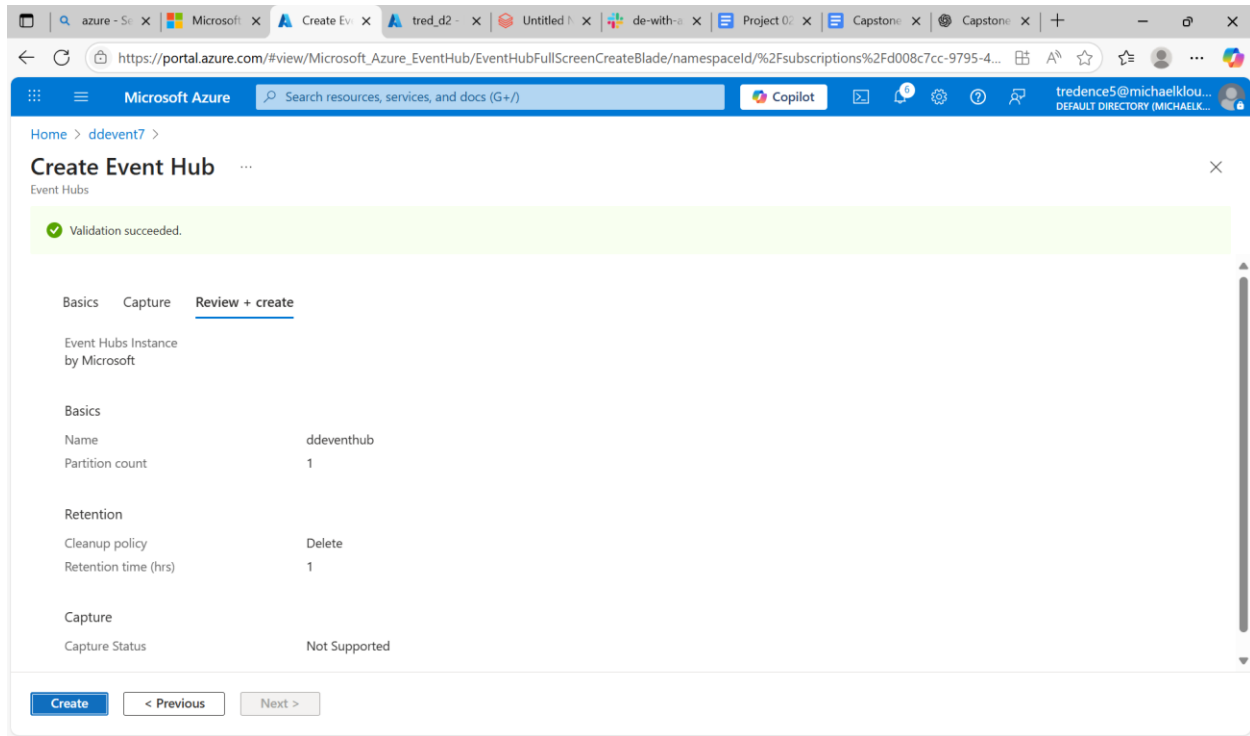
Secure your apps and infrastructure [Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials

[Start learning today >](#)

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support. [Find an Azure expert >](#)



Generating shared access policies

This screenshot shows the Microsoft Azure portal interface. The main page displays the 'ddeventhub (ddevent7/ddeventhub)' Event Hubs Instance. The left sidebar contains navigation options: Overview, Access control (IAM), Diagnose and solve problems, Data Explorer, Resource visualizer, Settings, and Shared access policies. The 'Shared access policies' section is selected, showing a table with no policies yet. On the right, the 'Add SAS Policy' dialog is open. It includes a 'Policy name' field with the value 'ddpolicy', a 'Manage' checkbox checked, and 'Send' and 'Listen' checkboxes also checked. A 'Create' button is at the bottom right.

This screenshot shows the 'SAS Policy: ddpolicy' configuration page in the Microsoft Azure portal. The page displays the 'ddpolicy' policy under the 'ddeventhub (ddevent7/ddeventhub)' Event Hubs Instance. The left sidebar is the same as the previous screenshot. The right panel shows the policy configuration, including options to 'Regenerate primary key', 'Regenerate secondary key', or 'Delete'. It also shows the 'Event Hubs access control with Shared Access Signatures' link. The 'Claims' section has 'Manage', 'Send', and 'Listen' checkboxes checked. Below this, there are fields for 'Primary key', 'Secondary key', 'Primary connection string', and 'Secondary connection string', each with a copy icon. At the bottom, the 'SAS policy ARM ID' is displayed, followed by 'Save' and 'Discard changes' buttons, and a 'Give feedback' link.

```
# Event Hub connection parameters
```

```
eh_namespace = "ddevent7"
```

```
eh_name = "ddeventhub"
```

```
eh_sas_key_name = "ddpolicy"
```

```
eh_sas_key = "77rnrrVErcZHLhcCfuvB1/C/tlsxG324i+AEhEFCgss="
```

```
connection_string =
```

```
f"Endpoint=sb://ddevent7.servicebus.windows.net/;SharedAccessKeyName=ddpolicy;SharedAccessKey=77rnrrVErcZHLhcCfuvB1/C/tlsxG324i+AEhEFCgss=;EntityPath=ddeventhub"
```

```
from pyspark.sql.types import StructType, StructField, StringType, DoubleType, TimestampType, IntegerType
```

```
from pyspark.sql import functions as F
```

```
# Define schema for Event Hub messages
```

```
event_schema = StructType([
```

```
StructField("OrderID", StringType(), True),
```

```
StructField("CustomerID", StringType(), True),
```

```
StructField("Region", StringType(), True),
```

```
StructField("StoreID", StringType(), True),
```

```
StructField("Product", StringType(), True),
```

```
StructField("Quantity", IntegerType(), True),
```

```
StructField("UnitPrice", DoubleType(), True),
```

```
StructField("SalesAmount", DoubleType(), True),
```

```
StructField("OrderTime", StringType(), True), # can cast to Timestamp later in Silver
```



```

StructField("DeviceType", StringType(), True),
StructField("Temperature", DoubleType(), True),
StructField("DeviceStatus", StringType(), True),
StructField("AnomalyFlag", IntegerType(), True),
StructField("AnomalyType", StringType(), True)
])

# Read streaming data from Event Hub

stream_df = (spark.readStream

.format("eventhubs")

.option("eventhubs.connectionString", connection_string)

.load()

)

# Event Hub messages are in 'body' column as binary, decode them

decoded_df = (stream_df

.withColumn("body_str", F.col("body").cast("string"))

.select(F.from_json("body_str", event_schema).alias("data"))

.select("data.*")

)

# Write stream to Bronze Delta table

bronze_stream = (decoded_df.writeStream

.format("delta")

.outputMode("append")

.option("checkpointLocation", "/mnt/delta/events_bronze/_checkpoints/")

.table("ddstorage77.bronze.events_bronze") # new bronze table for streaming

```

)

We get an error because we don't have admin access

The screenshot shows the Databricks workspace interface. The left sidebar contains navigation options: New, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Job Runs, Data Ingestion, and AI/ML. The main area displays a notebook titled 'Untitled Notebook 2025-08-25 11:27:13'. The notebook code is as follows:

```
25 .option("eventhubs.connectionString", connection_string)
26 .load()
27 )
28 # Event Hub messages are in 'body' column as binary, decode them
29 decoded_df = (stream_df
30   .withColumn("body_str", F.col("body").cast("string"))
31   .select(F.from_json("body_str", event_schema).alias("data"))
32   .select("data.*"))
33 )
34 # Write stream to Bronze Delta table
35 bronze_stream = (decoded_df.writeStream
36   .format("delta")
37   .outputMode("append")
38   .option("checkpointLocation", "/mnt/delta/events_bronze/_checkpoints/")
39   .table("ddstorage77.bronze.events_bronze") # new bronze table for streaming
40 )
41 )
```

The execution failed with the error: `[UNSUPPORTED_STREAMING_SOURCE_PERMISSION_ENFORCED] Data source eventhubs is not supported as a streaming source on a shared cluster. SQLSTATE: 0A000`. The error message is displayed in a red box. Below the error message are buttons for 'Diagnose error' and 'Debug'. The status bar at the bottom shows the temperature as 75°F, mostly cloudy, and the time as 10:41 AM on 8/26/2025.